Sahaj Bhadja

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Summary

Forward Deployed Software Engineer with 2+ years of experience delivering **Python data platforms**, **cloud architecture**, large-scale integrations, and **full-stack applications** from design through production.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Forward Deployed Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **data validation and reconciliation**, error handling, cloud storage, CI/CD releases, and production monitoring supporting dependable operational workflows and internal users reliably.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Increased pipeline efficiency by 68% through **AWS Glue and S3 ETL**, transforming JSON, CSV, and Parquet inputs into normalized datasets while reducing storage costs by 15% reliably.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform**, defining reusable components for customers, billing, allocation, permissions, and notifications with real-time updates across distributed locations reliably.
- Reduced resource wastage by 42% through **automated testing and deployment**, implementing JWT access controls, WebSocket updates, Azure monitoring, Docker/Jenkins pipelines, and ARM templates across eight modular service domains.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Delivered **sub-200ms REST APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.

Skills

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, Bash
Data & Analytics: ETL, Data Pipelines, Data Modeling, Schema Design, Query Optimization, Data Validation, PostgreSQL, MySQL, MongoDB, AWS RDS, Azure SQL, Firebase
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, Microservices, JSON, JWT
Frontend: React, HTML, CSS, Responsive Design, Reusable Components, Dashboard Development
Cloud & Infrastructure: AWS, Azure, GCP, AWS Glue, Amazon S3, Docker, Kubernetes, Jenkins, CI/CD, Cloud Storage
Engineering & AI: Object-Oriented Programming, Data Structures, Algorithms, System Design, LLMs, LLaMA-2, FAISS, Pinecone, Vector Search, Git, GitHub, Unit Testing, Integration Testing, Automated Testing, Debugging, Monitoring, Error Handling, Fault Isolation, Technical Documentation, Agile, Scrum

Projects

Enterprise Workflow Automation Platform - Design Concept
- Designed an **enterprise workflow-automation platform** concept connecting APIs, databases, approval steps, role-based access, configurable transformations, operational dashboards, deployment controls, and traceable execution records across legacy business processes.

Chat2SQL | Live
- Extended a **Java and TypeScript database workspace** supporting 10+ database systems, using Docker, Maven, Node.js, and Yarn for cross-platform schema exploration, query workflows, local debugging, and deployment.

Big-Brain | Live
- Built a **cloud-based research platform** combining REST APIs, PostgreSQL, vector search, role-based access, and retrieval workflows to organize heterogeneous information and deliver searchable, user-focused knowledge experiences reliably.